RNO1 – RELEVANT EXTRACTION AND CONTRACTION OF LOCAL INTERACTIONS

Abstract. This focused working outline separates relevant-term extraction, bounded reconstruction and normalized contraction for the supplied chamber activity space. It states the coefficient norms and field domains once, isolates the small-polymer analytic-disk argument, and gives linked statements, source locations and proof estimates.

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1. Introduction and target statement

RNO1 is the third multiscale proof project within QEO1. Its main estimate is linear in the incoming activity. Nonlinear fluctuation integration is the next project.
Current proof manuscript: RNM1, Relevant extraction and contraction of local interactions.
Overall plan. QEO1: the full construction.
Source editions. The 263-page master and its TeX source are the preserved comprehensive edition of QEM1. Its page, section and line locators below refer to that edition. Current companion manuscripts are cited by their codes and result names.

Original inputs: S1, curvature limits and the fourth cumulant; S2, chamber fermions and reflection positivity; S3, chamber renormalization and the fourth cumulant.

T. Extraction, reconstruction and normalized contraction

Model-specific. Overview Target.

Setting and hypotheses. Use the activity algebra, norm and independent connection domains $\mathcal{A}_k^{x,\mathbb{C}}$, $\widetilde{R}_k$, and $\mathfrak{X}_k$ defined in D. The norm counts ordinary and covariant-difference generators separately, with weights $h_1 = e_k^{-1/4}$ and $h_2 = e_k^{-1/4+\varepsilon}$. Let $L \geq 5$ be odd, $M = L^m \geq L^2$, and $\eta = L^{-k}$ for a nonnegative integer k, and assume that both $\ell/(LM)$ and $\ell/(L^2M)$ are odd integers at least 13. Set

$$\bar{\varepsilon} = 1/200, \quad c_X = 54, \quad \kappa_0 = 54 \log 72, \quad K_0 = e^{\kappa_0}/36, \quad c_{\mathrm{rb}} = c_X K_0, \quad c_{\mathrm{fl}} = 61 c_{\mathrm{rb}}.$$

The contraction theorem has the following explicit parameter hypotheses:

$$0 < \varepsilon \leq \bar{\varepsilon}, \quad 1 - \varepsilon < \alpha < 1, \quad L^{1/8 - 3\bar{\varepsilon}/2} \geq 2400, \quad e_k^{\varepsilon} \leq 1/4,$$
$$160 L^{1+3\varepsilon/2} M e_k^{\varepsilon} \leq 1, \quad 9 M L^{\varepsilon} e_k^{2\varepsilon} \leq 1, \quad 90 c_X L M e_k^{\varepsilon} \leq 1,$$
$$\kappa \geq 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \geq \kappa.$$

Take $E \in \mathcal{A}_k^{x,\mathbb{C}}$.
Principal result: extraction and contraction. There are linear vacuum and mass functionals, real on real activities, with

$$|\varepsilon_v(E)| \le K_0 M^{-3} \|E\|_k, \qquad |m(E)| \le K_0 M^{-3} e_k^{1/2} \|E\|_k.$$

The extracted mass matrix is $m(E)1_4$. After subtracting these terms and applying the prescribed shortest-chain adjustment, the activity $E_{\mathrm{n}} = (RE)^{\mathrm{nat}}$ has zero vacuum and zero ordinary quadratic kernel at zero connection on each small polymer. The scalar extraction is exact before pair evaluation, and adjustment preserves the evaluated interaction.

Set $e_{k+1} = \sqrt{L} e_k$. Rescale ordinary fields by L^{-1}, pair fields by $L^{-1-\alpha}$, and connections by $B_L(b) = L^{-1/2} B(b/L)$, and sum fine polymers with the same LM-cube closure. This defines the linear reblocking map $\mathcal{L}$ of S11. It preserves the declared symmetries, reality and restriction to zero external connection, and

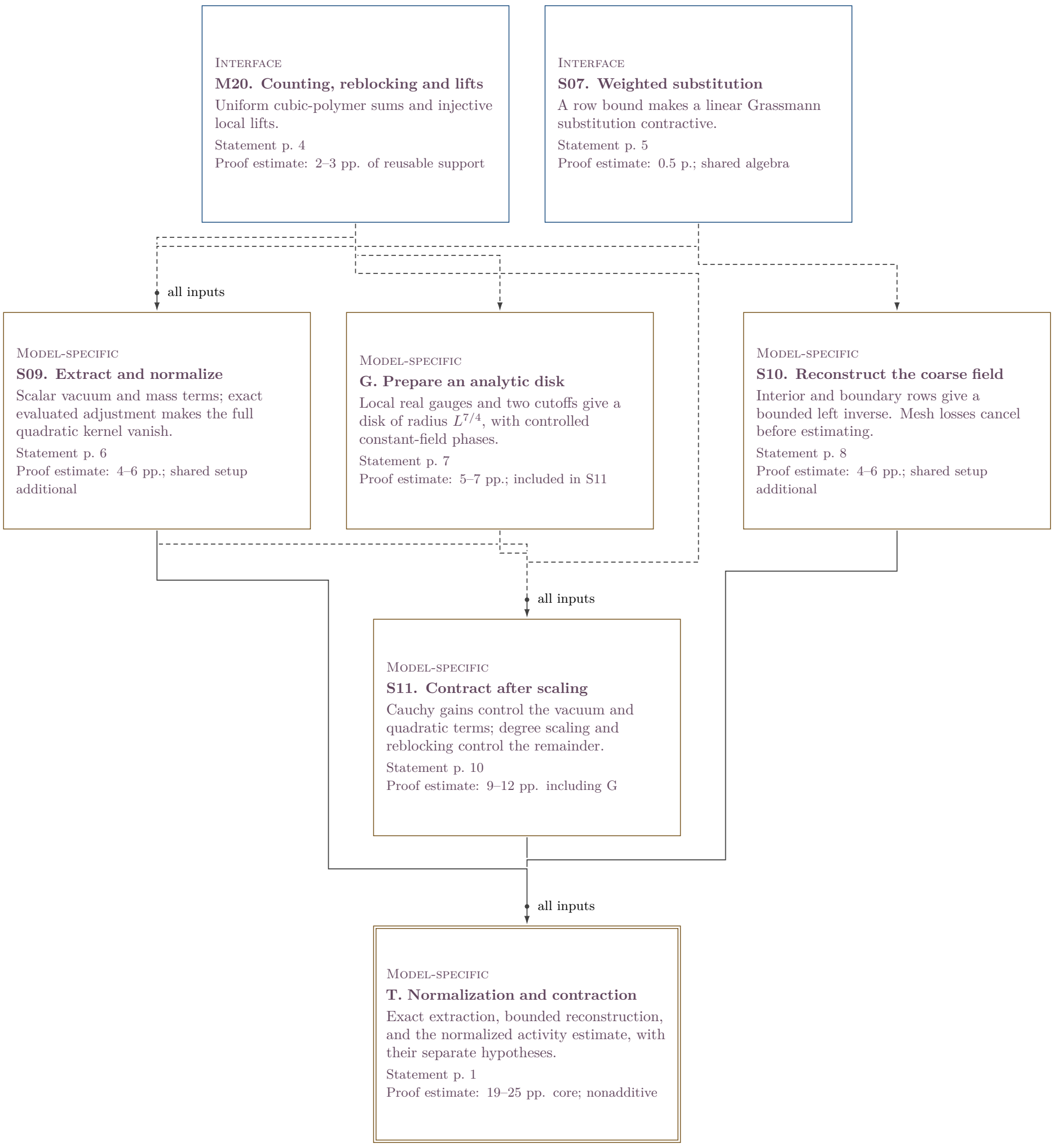
$$\|\mathcal{L} E_{\mathrm{n}}\|_{k+1} \le c_{\text{fl}} L^{-1/4+3\varepsilon/2} \|E\|_k.$$

For $w_k^E = e_k^{1/4-7\varepsilon}$, the relative factor is $c_{\text{fl}} L^{-3/8+11\varepsilon/2}$. It is at most $1/16$ under the additional choice $c_{\text{fl}} L^{-3/8+6\bar{\varepsilon}} \le 1/16$.
Separate result: bounded reconstruction. Under the chamber-stencil, phase and invertibility hypotheses of S10, the coarse-to-fine mean map has a left inverse represented using ordinary interior fields and boundary pair fields. The row bound is independent of mesh and polymer. With the stated weight inequality this substitution does not increase the coefficient norm and returns the original coarse field after evaluation. This result is an additional output; it is not a hypothesis of the contraction assertion.
Proof organization. The low-degree terms prevent contraction by naive scaling. Scalar extraction removes the vacuum and integrated mass. The shortest-chain adjustment makes the entire ordinary quadratic kernel vanish. On small polymers, G then supplies an analytic disk large enough to gain an additional factor in precisely these two degrees. All other degrees contract by scaling; large polymers gain from their exponential weight. M20 controls the reblocking sum. Reconstruction has a different central estimate: cancellation in the averaged chamber stencil removes an apparent inverse-mesh loss.
Source and budget. 263-page master, §12, pp. 113–133, and the reconstruction substitution in §13.4. The supporting geometry is Appendix A. Budget 19–25 pages for the principal extraction, reconstruction and contraction proofs, with their common setting written once. Supporting geometry and weighted algebra are recalled explicitly; their proof budgets must not be added again if supplied elsewhere. The current proof manuscript is RNM1. The ranges above are the original planning estimates.

RNO1 – Relevant extraction and contraction of local interactions

Definitions/setup Model-specific Interface (general)

Focused outline within QEO1. The common activity space and all standing parameter conditions are stated in D and T. Reconstruction and contraction are separate outputs.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

All results remain conditional on their stated domains and numerical inequalities. S10 uses the explicitly recalled chamber stencil and mean map. It is not an input to S11. The estimate for G is included in S11, and shared setup and imported support are counted once.

2. M20. Polymer counting, reblocking and nonwrapping lifts

Interface. Overview Target.

Objects and hypotheses. Let $L \geq 5$ be odd, $M \geq L^2$ a power of L, and let a cubic torus side ℓ have $\ell/(LM)$ an odd integer at least 13. A polymer X is a nonempty face-connected union of M-cubes. Define $Md_M(X)$ as the infimum length of a connected net meeting their closed cubes. A net is a union of images of specified straight segments; length is the sum of their represented Euclidean lengths, with the infimum also over those representations. Let $\overline{X}^{\,L}$ be its closure in LM-cubes.
Claim. With $c_X = 54$, $\kappa_0 = 54 \log 72$, $K_0 = e^{\kappa_0}/36$, prove, with $|X|_M$ the number of cubes,

$$d_M(X) \leq |X|_M - 1, \quad |X|_M \leq c_X(d_M(X)+1), \quad d(p,q) \leq M(d_M(X)+2) \quad (p,q \in X),$$
$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \leq K_0, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \leq K_0 |Y|_M,$$

where sums run over nonempty connected polymers and Y is any cube union. For reblocking require also that ℓ/L has the same side conditions, and prove

$$Ld_M(Y) \leq d_M(X) \quad (\overline{X}^{\,L} = LY), \qquad \sum_{\overline{X}^{\,L} = LY} e^{-\kappa' d_M(X)} \leq c_X K_0 L^3 e^{-L(\kappa' - \kappa_0 - 1) d_M(Y)}.$$

Here $\kappa' \geq \kappa_0$. If $2M(d_M(X)+2) < \ell$, construct an injective adjacency-preserving covering lift $\widetilde{X}$ with the same net distance. If also $M(d_M(X)+4) < \ell$ and $M \geq 2$, then at every mesh $\eta = L^{-k} \leq 1$ there is a lattice box W containing every bond meeting $\widetilde{X}$, with each side at most $M(d_M(X)+3)$. Projection is injective on W, and every torus bond meeting X has exactly one lift in W. Construct W by enlarging each coordinate interval of the closed lifted cubes by one mesh at either end.
Proof work. Construct a rooted cube tree from a near-minimal represented net, count its finite neighbours and tree codes, and integrate the exponential tail. Scale the represented segments before taking the infimum for reblocking. A path shorter than one period cannot identify distinct required lifts; check every adjacency and the enlarged bond box separately. All constants must remain independent of torus side.
Inputs. Cubic geometry only. This is an upstream geometric lemma used by localization, cover comparison and polymer logarithms; it has no field-theoretic input.
Source. Manuscript pp. 261–263, Appendix A and Proposition A.1.

Supporting proof: 2–3 pages if not imported; source Appendix A is reused.

3. S07. Weighted substitution in a finite Grassmann algebra

Interface. Overview Target.

Statement. Let I_i and J_j be finite ordered sets with positive index measures ν_i, ν'_j. Give generators in the input family i weight $w_i > 0$, and generators in the output family j weight $w'_j > 0$. The coefficient norm is the sum of absolute ordered-monomial coefficients times their generator weights. Equivalently, write antisymmetric kernels with the index measures and factorials. This norm is submultiplicative. The linear substitution

$$\varphi_i(t) \longmapsto \sum_j \sum_{s \in J_j} O_{ij}(t,s)\nu'_j(s)\theta_j(s)$$

extends to an algebra homomorphism of norm at most one if, for every i,

$$\sum_j \left(\sup_{t \in I_i} \sum_{s \in J_j} |O_{ij}(t,s)|\nu'_j(s) \right) w'_j \le w_i.$$

External generators may be retained as additional families.

Use and proof task. Apply the triangle inequality to each image generator, then use submultiplicativity on ordered monomials. This controls the chain adjustment and the reconstruction substitution with the actual ordinary and pair weights. The factors $\eta^3/48$ and $(\eta^3/48)^2$ belong to the kernels, not to generator weights.

Source and budget. This is the substitution clause of QEO1 S07, from 263-page master §11.4, pp. 107–110, Proposition 11.4. Allow about half a page when the coefficient norm has already been defined. The Gaussian-integration clauses of S07 are not needed here.

4. S09. Scalar extraction and exact quadratic normalization

Model-specific. Overview Target.

Setting. Use the complete activity space of D, the counting and lift bounds M20, and weighted substitution S07. The extraction assertion needs $\kappa \geq \kappa_0$. The adjustment additionally needs $90c_X LMe_k^{\varepsilon} \leq 1$ and $e_k^{\varepsilon} \leq 1/4$.

Claim. Write $E^{[2,0]} = \int \bar{\psi}(x)^T K_X(x,x')\psi(x')\,\mathrm{d}x\,\mathrm{d}x'$. On small X, define

$$a_0 = \frac{E_{00}(X;0,0)}{\mathrm{Vol}(X)}, \qquad a_2 = \frac{1}{\mathrm{Vol}(X)} \int_{\widetilde{X}\times\widetilde{X}} K_X(0,0;x,x').$$

The sums $\varepsilon_v(E) = -\sum_{X\supset\square,\ \mathrm{small}} a_0$ and $M(E) = -\sum a_2$ are independent of the cube. In fact $M(E) = m(E)1_4$, with complex-linear maps real on real activities, and

$$|\varepsilon_v(E)| \leq K_0 M^{-3}\|E\|_k, \qquad |m(E)| \leq K_0 M^{-3} e_k^{1/2}\|E\|_k.$$

Subtract $a_0\,\mathrm{Vol}(X)$ and $\int_X \bar{\psi}^T a_2 \psi$ only on small polymers, to obtain RE. Then $\|RE\|_k \leq 2\|E\|_k$ and

$$\sum_X E(X) = -\varepsilon_v(E)\ell^3 - m(E)\langle\bar{\psi},\psi\rangle + \sum_X RE(X).$$

There is an activity adjustment, preserving evaluated values. Applied to RE, it makes the entire ordinary quadratic kernel vanish at $(A,C) = (0,0)$ on each small polymer. Construct it by averaging over shortest face-chains between the cubes of $x, \widetilde{x}$, retaining the initial chamber label at intermediate cube centres, and telescoping

$$\psi(p_i) = U_{A+C}(p_i,p_{i+1})\psi(p_{i+1}) - d_X(p_i,p_{i+1})^{\alpha}\chi(p_i,p_{i+1}).$$

Average over $\widetilde{x}$ with normalized measure $\mathrm{Vol}(X)^{-1}\mathrm{d}\widetilde{x}$; use inverse transports for barred legs. This defines a homomorphism Θ. Set $E^{\mathrm{nat}} = E - E^{[2,0]} + \Theta E^{[2,0]}$. If $90(54)LMe_k^{\varepsilon} \leq 1$ and $e_k^{\varepsilon} \leq 1/4$, then its pointwise norm is at most $10\|E(X)\|$. For $(RE)^{\mathrm{nat}}$, the vacuum and full ordinary quadratic kernel are both zero at zero connection on small X. All these maps preserve the activity symmetries and commute with setting the external slot to zero.

Proof work. Derive matrix covariance from unique antisymmetric kernels. The summed extraction commutes with all four matrices $\gamma_1,\gamma_2,\gamma_3,\gamma_5$, whose commutant is scalar. Use the quadratic weight $h_1^{-2} = e_k^{1/2}$ and polymer counting. Reorder finite sums for the exact extraction identity. For adjustment keep the chain length and phase products: the path length and sum of d_X^{α} are at most $3(54)LM$. Telescoping proves equality only after the prescribed field evaluation. At zero connection the remaining ordinary part is a common spatial mean, annihilated by the zero integrated kernel. Use $h_2/h_1 = e_k^{\varepsilon}$ for the norm bound.

Inputs. Weighted Grassmann substitutions; explicit activity symmetries; polymer counting/lifts. Exports: scalar counterterms and a zero ordinary quadratic kernel. Individual polymer matrices a_2 need not be scalar.

Source. PDF pp. 114–118, 122–124, §§12.1–12.3,12.5, Proposition 12.1, (12.1)–(12.17), (12.34)–(12.39). TeX 10142–10565, 10905–11070.

Proof estimate: 4–6 pages, within the 10–13-page extraction/reconstruction assignment.

5. G. Small-polymer gauges and the analytic disk

Model-specific. Overview Target.

Setting and hypotheses. Use D and the contraction parameter hypotheses of T. Let X be small, meaning $d_M(X) \le L$, and take a coarse connection pair $(A, C) \in \widetilde{R}_{k+1} \times \mathfrak{X}_{k+1}$. Put $\theta_i = e_k^{-3/4+i\varepsilon}$, $d_1 = e_k^{-1+2\varepsilon}$, and $d_2 = e_k^{-1+3\varepsilon}$.

Statement. There are a real function λ_*, constant bond fields K_1, K_2, and global bond fields A^b, C^b such that on all bonds with an endpoint within distance one of X,

$$A_L - d\lambda_* = A^b + K_1, \qquad C_L = C^b + K_2.$$

The respective three norm-to-threshold ratios of A^b are at most

$$(64, 137, 1382) L^{-15/8+3\varepsilon/2},$$

and both ratios of C^b are less than $L^{-7/4}$. Moreover

$$\|K_1\|_\infty < L^{-7/8+\varepsilon/2}\theta_1, \qquad \|K_2\|_\infty < L^{-1+\varepsilon} d_1.$$

In particular $(tA^b, tC^b) \in R_k \times \mathfrak{X}_k$ for $|t| \le L^{7/4}$. When $C = 0$, the construction gives $C^b = K_2 = 0$.

For any local activity $F(X)$ holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, with the support and real-gauge covariance in D, the corresponding kernels satisfy

$$|F_{nm}(X; A_L, C_L; \xi; \zeta)| \le \exp(2(n+m))\, |F_{nm}(X; A^b, C^b; \xi; \zeta)|.$$

The vacuum kernels are equal.

Main argument to write. Lift the small polymer and choose a base point. Remove the real symmetric linear part of a local real representative by a real quadratic gauge. Cut off the Taylor remainder at radius $2LM$. The remaining real curvature and complex linear part need a second radius, $\min(e_k^{-\varepsilon}, \ell/5)$; their slopes are bounded by $2\min(b_2, \ell b_3)$, where $b_2 = L^{-15/8+\varepsilon}\theta_2$ and $b_3 = L^{-15/8+3\varepsilon/2}\theta_3$. These two cutoffs give the displayed global norms. Continue the real constant-gauge identity to complex constants on its explicitly allowed connected domain. The bounded imaginary phase at each leg gives the kernel comparison. An assumed analytic disk would omit the principal argument.

Source and budget. 263-page master §12.7, Propositions 12.3–12.4, equations (12.48)–(12.61), TeX 11214–11601 in the frozen source. Allow 5–7 pages within the 9–12-page contraction assignment. The two-cutoff construction and the phase bound deserve the space.

6. S10. A bounded left inverse using interior and boundary legs

Model-specific. Overview Target.

Additional reconstruction setting. At positive level k, let $\eta = L^{-k}$, $0 < \mu \le 1$, $b_k \ge b_- > 0$. On chamber spinors with measure $\eta^3/48$, $\mathcal{Q} = Q_k(A)J^*$ is the normalized nested block average, $\mathcal{Q}^\sharp = JQ_k(-A)^T$, $\mathcal{Q}\mathcal{Q}^\sharp = 1$. The chamber stencil is $D_A = -\eta^{-1}\sum_i M_i(x)(U_A R_i - 1)$, with three internal and one external neighbor, $M_i = \frac{1}{2}w_i(\rho - \gamma\cdot\widehat{d}_i)$. Here $0 < \rho < 1$, $w_0 = w_3 = 6, w_1 = w_2 = 6\sqrt{2}$, and the directions are $\widehat{d}_0 = -ge_3$, $\widehat{d}_1 = -g(e_2 - e_3)/\sqrt{2}$, $\widehat{d}_2 = -g(e_1 - e_2)/\sqrt{2}$, $\widehat{d}_3 = ge_1$, with neighbours as in D. The block phases are the nested six-path averages described below. Its geometric cancellation is $\sum_i w_i\widehat{d}_i = 0$; the external matrices have norm at most 6. Let

$$F_\infty = D_A + \mu + b_k\mathcal{Q}^\sharp\mathcal{Q}, \quad H_k = b_k F_\infty^{-1}\mathcal{Q}^\sharp, \quad T_k = \mathcal{Q}(b_k^{-1}D_A + 1 + \mu/b_k).$$

The inverse is assumed only where $T_kH_k = 1$ is used. Write $c_3 = c + \eta g e_1$ for the photon point of $x^{(3)}$. For a union Y of partition M-cubes, split the row of T_k at a unit site $y \in Y$ into $T_Y^{(1)}$, retaining ordinary differences whose neighbor stays in Y, and

$$(T_Y^{(2)}\chi)(y) = -b_k^{-1}\eta^{\alpha-1}\int_{\Delta_y}\mathbf{1}[c_3 \notin Y]\phi_A(y,c)M_3(x)\chi(x,x^{(3)})\,\mathrm{d}x.$$

The nested phase has modulus at most $\exp(3\mathfrak{i}/2)$, where $\mathfrak{i} = e_k\|\operatorname{Im}A\|_\infty$; put $q_d = e_k\|dA\|_\infty$.

Averaging convention. Δ_y is the unit cube centred at $y \in \mathbb{Z}^3/\ell\mathbb{Z}^3$. Starting from a photon site, join it to the successive centres of its nested L-adic blocks. At each stage use the arithmetic mean of the six coordinate-order path sums. Their total exponential is $\phi_A(y,c)$. Then

$$(\mathcal{Q}v)(y) = \frac{\eta^3}{48}\sum_{c\in\Delta_y,g}\phi_A(y,c)v(c,g), \qquad (\mathcal{Q}^\sharp v)(c,g) = \phi_A(y,c)^{-1}v(y).$$

Each path sum contains ηe_k. The superscript T denotes the weighted algebraic transpose. The reverse map $\mathcal{Q}^\sharp$ is obtained by transposing the average at the reversed connection; these are distinct operations.

Claim. $T_kH_k = 1$. On evaluated boundary legs $\chi = \delta_{\alpha,A}\psi$, $T_Y^{(1)}\psi + T_Y^{(2)}\chi = T_k\psi$ on Y. Only ordinary legs over Y and external boundary pairs occur; coefficients read bonds meeting Y. Uniformly in the mesh and Y,

$$\|T_Y^{(1)}\|, \|T_k\| \le e^{3\mathfrak{i}/2}[1 + b_-^{-1}(54Lq_de^{6q_d} + 12 + 6e^{\mathfrak{i}} + \mu)],$$
$$\|T_Y^{(2)}\| \le 6b_-^{-1}e^{3\mathfrak{i}/2}\eta^\alpha.$$

These are Euclidean spinor sup norms; the scalar coefficient row sum is at most twice this bound. If $\mathfrak{i}, q_d \le 1$, all ordinary/barred rows are bounded by

$$C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)].$$

They have the required lattice, local real-gauge, charge and reality covariance. The substitution $\Psi \mapsto T_Y^{(1)}\psi + T_Y^{(2)}\chi$, with its barred counterpart, preserves support and decreases coefficient norm if the coefficient-norm weights $\omega_1, \omega_2, \omega_3$ of the ordinary, pair and unit generators ψ, χ, Ψ satisfy $C_T(\omega_1 + \omega_2) \le \omega_3$. On evaluating $\psi = H_k\Phi$ and its covariant pair field, the substitution returns Φ on Y. At zero level these maps are identity on all chamber components.

Proof work. The left inverse is algebraic, but its uniform row estimate requires the cancellation of the apparent η^{-1} loss. Reindex internal hops and internal external hops, using $M_i(x^{(i)}) - M_i(x) = w_i\gamma \cdot \widehat{d}_i(x)$. Boundary site counts are $48\eta^{-2}$, compared with $48\eta^{-3}$ total sites in a unit block. For the remaining nested-phase difference, classify a site by the first common block scale j. There are at most $\eta^{-3}L^{-(j-1)}$ sites, while the closed-path phase error is at most $6L^{2(j-k)}q_d \exp(6q_d)$. Sum these factors before taking a bound. The boundary Hölder row gains η^{α} from its face count. Verify the homomorphism and covariance on each independent generator.

Inputs. Exact field map and nested gauge averages; chamber cancellation; weighted substitution theorem. Exports: support-preserving reconstruction and its explicit uniform row constant. The activity keeps pair generators independent until evaluation.

Source. PDF pp. 118–122, 139–140, §§12.4,13.4, Proposition 12.2, (12.18)–(12.33), (13.23)–(13.25). TeX 10566–10904, 12768–12866.

Proof estimate: 4–6 pages, within extraction and reconstruction.

7. S11. Scaling and contraction of a normalized local activity

Model-specific. Overview Target.

Objects and hypotheses. The activity input is exactly $\mathcal{A}_k^{x,\mathbb{C}}$ defined in D, with its complete kernel support, measures, analytic domains and three covariance equations; for reality use the additional antilinear equation defining $\mathcal{A}_k^x$. The local description below uses this fixed package. Let L be odd, $M = L^m \geq L^2$, $\eta = L^{-k}$, and both $\ell, \ell/L$ satisfy the admissible side condition $\ell/(LM)$ odd and at least 13. A polymer is small when $d_M(X) \leq L$. Assume its nonwrapping lift, size/counting bounds with constants $c_X = 54, \kappa_0, K_0$, and

$$\sum_{\overline{X}^L = LY} e^{-\kappa' d_M(X)} \leq c_{\mathrm{rb}} L^3 e^{-L(\kappa' - \kappa_0 - 1) d_M(Y)}, \quad c_{\mathrm{rb}} = c_X K_0, \quad \kappa' \geq \kappa_0.$$

An activity has ordinary legs over X and pair legs either with all six rectilinear paths in X, or external boundary pairs with an endpoint there. Weights are $h_1 = e_k^{-1/4}, h_2 = e_k^{-1/4+\varepsilon}$, and its norm is the supremum of the weighted coefficient norm multiplied by $e^{\kappa d_M(X)}$. Assume balance, holomorphy, local read dependence, independent local real gauges of its two slots, and charge conjugation, so the vacuum is even under $(A, C) \mapsto (-A, -C)$. For the stated preservation conclusions also retain the lattice and reality actions of the activity. The first slot has three thresholds $\theta_i = e_k^{-3/4+i\varepsilon}$, allowing local real gauges plus a global complex part; the external slot has thresholds $d_1 = e_k^{-1+2\varepsilon}$, $d_2 = e_k^{-1+3\varepsilon}$. Use

$$0 < \varepsilon \leq \bar{\varepsilon} = 1/200, \quad 1 - \varepsilon < \alpha < 1, \quad L^{1/8 - 3\bar{\varepsilon}/2} \geq 2400, \quad e_k^{\varepsilon} \leq 1/4,$$
$$160 L^{1+3\varepsilon/2} M e_k^{\varepsilon} \leq 1, \quad 9 M L^{\varepsilon} e_k^{2\varepsilon} \leq 1, \quad 90 c_X L M e_k^{\varepsilon} \leq 1,$$
$$\kappa \geq 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \geq \kappa.$$

For an input E, let $E_{\mathrm{n}} = (RE)^{\mathrm{nat}}$: its small-polymer vacuum and ordinary quadratic kernel are zero at zero connection; pointwise it has norm at most 20 times the original norm, unchanged on large polymers.

Claim. Set $e_{k+1} = \sqrt{L} e_k$, $A_L = L^{-1/2} A(\cdot/L)$, and

$$S_L \psi(x) = L^{-1} \psi'(x/L), \quad S_L \chi(x, x') = L^{-1-\alpha} \chi'(x/L, x'/L), \quad (\mathcal{L}F)(Y) = \sum_{\overline{X}^L = LY} S_L F(X; A_L, C_L).$$

Here $\overline{X}^L$ is the union of LM-cubes containing X. On bidegree (n, m),

$$\|(S_L F)^{[n,m]}\|_{h_{k+1}} = L^{-p(n,m)} \|F^{[n,m]}\|_{h_k}, \quad p(n,m) = n + (1+\alpha)m + (n+m)/8 - m\varepsilon/2.$$

The scaled pair range is M/L, including intracluster pairs. The linear normalized map preserves the activity symmetries and the zero-external-slot restriction, and

$$\|\mathcal{L} E_{\mathrm{n}}\|_{k+1} \leq c_{\mathrm{fl}} L^{-1/4+3\varepsilon/2} \|E\|_k, \qquad c_{\mathrm{fl}} = 61 c_{\mathrm{rb}}.$$

Relative to $w_k^E = e_k^{1/4-7\varepsilon}$, its factor is $c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2}$; it is at most $1/16$ if $c_{\mathrm{fl}} L^{-3/8+6\bar{\varepsilon}} \leq 1/16$.

Proof work. For a small polymer, use the representatives A^b, C^b from G, whose analytic disk has radius $\varrho = L^{7/4}$ and whose kernel comparison costs $\exp(2(n+m))$. Put

$$N(X) = \sup_{(A,C) \in \widetilde{R}_k \times \mathfrak{X}_k} \|E_{\mathrm{n}}(X; A, C)\|_{h_k}, \qquad f_{nm}(t) = (E_{\mathrm{n}})_{nm}(X; tA^b, tC^b).$$

The factorial-weighted coefficient norm of $f_{nm}(t)$ is at most $N(X)$ for $|t| \leq \varrho$. Normalization gives $f_{00}(0) = 0$; charge conjugation makes the vacuum even, so $f'_{00}(0) = 0$. The second-order Cauchy remainder is therefore bounded by $N(X)/[\varrho(\varrho - 1)] \leq 2L^{-7/2} N(X)$.

For the ordinary quadratic term, normalization gives $f_{20}(0) = 0$ as a full antisymmetric kernel. The Cauchy remainder has factor $1/(\varrho - 1)$. Combining it with the comparison factor $\exp(4)$ and

the degree-scaling factor $L^{-p(2,0)} = L^{-9/4}$ gives $2\exp(4)L^{-4}N(X)$. Other degrees gain $L^{-13/4+3\varepsilon/2}$. Reblock with the displayed geometry bound; large polymers supply the extra $\exp(-L)$. Finally prove that these formulas on covering generators commute with quotient maps under the margins (12.70),(12.74).

Inputs. Scalar extraction and quadratic normalization; weighted algebra; polymer reblocking/lifting.

Exports: contraction with explicit relative factor, scaled domains and lifted local formulas. Integrated quadratic cancellation alone does not give the quadratic Cauchy gain.

Source. PDF pp. 124–133, §§12.6–12.9, Propositions 12.3–12.5, (12.40)–(12.74). TeX 11071–12017.

Proof estimate: 9–12 pages including G; do not add G's estimate again.

Appendix A. Detailed proof tasks and supporting statements

D gives the common activity definition needed by the principal statements.

D. The activity space and its field domains

Setup. Overview Target.

Why this setting is shared. The extraction and contraction results concern the same finite Grassmann algebra. Its generators, coefficient norm and two connection domains are specified here once. The reconstruction result additionally uses the chamber stencil and nested averaging maps stated on its own page. None of these definitions requires an interacting renormalization trajectory.

Chamber labels and evaluation. A chamber site is $x = (c, g)$, where $c \in \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with $\eta = L^{-k}$ for a nonnegative integer k, and g ranges over the 48 signed permutation matrices in dimension three. There are four spin components and separate barred and unbarred generators. The three internal neighbours replace g by gs_i, where s_0 flips the third coordinate, s_1 exchanges the last two, and s_2 exchanges the first two. The external neighbour is $x^{(3)} = (c + \eta g e_1, g r_{e_1})$, where r_{e_1} flips the first coordinate. This neighbour convention fixes the external pairs used below. For a connection B, $U_B(x, x')$ is the exponential of ie_k times the arithmetic mean of its six coordinate-order path sums, each with the mesh factor η. It is one within a photon cluster. The pair generators remain independent until the evaluation

$$\chi(x, x') = \frac{U_{A+C}(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta)^\alpha}$$

and its barred version with inverse transport. Statements about an evaluated activity refer to this substitution.

The common setting. Define the activity space $\mathcal{A}_k^{x,\mathbb{C}}$, with real subspace $\mathcal{A}_k^x$. Fix odd $L \geq 5$, $\eta = L^{-k}$, $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13, $0 < \varepsilon \leq 1/200$, $0 < \alpha < 1$, and $0 < e_k < 1$. An M-polymer is small if $d_M(X) \leq L$, with an injective covering lift. Put $\mathrm{Vol}(X) = M^3|X|_M$. Balanced means that each monomial has equally many barred and unbarred generators. An activity $E(X; A, C)$ is a balanced polynomial in independent ordinary and pair Grassmann legs, with holomorphic coefficients reading only bonds meeting X. Ordinary legs are over X; an allowed pair has all six coordinate-order paths in X, or is an external nearest-neighbor pair with an endpoint over X. Its separation is $d_X = \max(|c - c'|_\infty, \eta)$, and the pair range is $0 < d_X \leq M$; intracluster pairs are included. Index measures are $\eta^3/48$ and $(\eta^3/48)^2$, while generator weights are $h_1 = e_k^{-1/4}$, $h_2 = e_k^{-1/4+\varepsilon}$. Ordinary kernels use the antiperiodic convention in each chamber argument; pair indices are simultaneous period classes of their two endpoints. The coefficient norm of the separately antisymmetric kernels is

$$\|E(X; A, C)\|_{h_k} = \sum_{n,m} \frac{h_1^n h_2^m}{n!m!} \int |E_{nm}(X; A, C; \xi; \zeta)| \, \mathrm{d}\xi \, \mathrm{d}\zeta.$$

To specify its domains, put

$$\begin{aligned} \partial_\nu B_\mu(c) &= \eta^{-1}[B_\mu(c + \eta e_\nu) - B_\mu(c)], \\ \|B\|_1 &= \sup |B_\mu(c)|, \quad \|B\|_2 = \sup |\partial_\nu B_\mu(c)|, \\ \|B\|_3 &= \sup_{\mu,\nu} \sup_{0<d(c,c')\leq 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c, c')^\alpha}, \qquad \theta_i = e_k^{-3/4+i\varepsilon}. \end{aligned}$$

The distance is periodic sup distance; a local norm uses only its contained bonds and difference cells. $R_k = \{B : \|B\|_i < \theta_i,\ i = 1, 2, 3\}$. A field belongs to $\tilde{R}_k$ if it is $A_0 + A_1$, A_0 real, $A_1 \in R_k$, and on every enlarged cube of side $(2L + 1)M$, $A_0 = A_\square + d\lambda_\square$ with real $A_\square, \lambda_\square$ and $\|A_\square\|_{i,\square^\dagger} < \theta_i$. The external domain is

$$\mathfrak{X}_k = \{C : \|C\|_1 < e_k^{-1+2\varepsilon},\ \|C\|_2 < e_k^{-1+3\varepsilon}\}.$$

The activity norm is precisely

$$\|E\|_k = \sup_X \exp(\kappa d_M(X)) \sup_{(A,C)\in \widetilde{R}_k\times\mathfrak{X}_k} \|E(X;A,C)\|_{h_k}.$$

Assume the polymer counting bound $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0$, $\kappa \ge \kappa_0$.

Here is the complete symmetry package. Use $\gamma_\mu = \sigma_3\otimes\sigma_\mu$, $\gamma_5 = \sigma_1\otimes 1_2$, $\mathsf{C} = i\gamma_2$, and $\varkappa = \gamma_5\mathsf{C}$. For a spin lift S generated by the four γ-matrices and a translation $a\in M\mathbb{Z}^3$, let $\widetilde{g}(c,g) = (r(S)c + a, r(S)g)$, where $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Define

$$a_{\widetilde{g}}\psi(x) = S\psi(\widetilde{g}^{-1}x), \qquad a_{\widetilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\widetilde{g}^{-1}x).$$

For pair generators transform both endpoints and use the same spin matrices. G_λ multiplies an unbarred ordinary or pair leg by $\exp(ie_k\lambda(c))$, with c its first photon point, and a barred leg by its inverse. Define on ordinary and pair legs

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \quad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \quad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

with $\mathbf{K}$ antilinear. The complex activity class requires

$$\begin{aligned} a_{\widetilde{g}}E(\widetilde{g}X;\widetilde{g}A,\widetilde{g}C) &= E(X;A,C),\\ \mathrm{Cg}\,E(X;-A,-C) &= E(X;A,C),\\ G_{\lambda+\lambda'}E(X;A^1,C^1) &= E(X;A,C) \end{aligned}$$

whenever the two pairs are in the domain and $A^1 = A - d\lambda$, $C^1 = C - d\lambda'$ on bonds at distance at most one from X, for real λ,λ'. Its real subspace additionally requires

$$\mathbf{K}E(X;-\overline{A},-\overline{C}) = E(X;A,C).$$

All equations act on independent generators before evaluation. For the reconstruction application in S10, the explicit conditions

$$e_k^{\varepsilon} \le \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^{\varepsilon} \le 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \le 1$$

give $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k$, where $\widetilde{G}_k$ has a global complex bound $e_k^{-1+\varepsilon}$ and local real-gauge sup bounds $e_k^{-1+\varepsilon}$ on every $11M$ cube. They also give $e_k\|\mathrm{Im}(A+C)\|_\infty \le 1$ and $e_k\|d(A+C)\|_\infty \le 1$.

Source. 263-page master §12.1; QEO1 S09. This is shared setup, not a separate estimate.